Project Summary: Design of a Wideband, High-Efficiency Pseudo-Doherty Load Modulated Balanced Amplifier

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Date: September 13, 2025

1. Project Overview

This report summarizes the comprehensive design and research process for a novel wideband, high-efficiency power amplifier (PA) intended for modern communication systems such as 5G. The primary objective was to address the critical challenge of maintaining high efficiency in PAs that handle signals with a high Peak-to-Average Power Ratio (PAPR). The project documents the systematic evolution of the design, beginning with the enhancement of a conventional Load Modulated Balanced Amplifier (LMBA) and culminating in the development of an advanced Pseudo-Doherty LMBA (PD_LMBA) that incorporates a novel matching network and is ready for fabrication.

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2. Design Progression and Key Technical Innovations

The research was conducted in four distinct phases, demonstrating a logical and iterative approach to complex RF engineering challenges.

- **Phase 1: Initial LMBA Enhancement**
  The project began by addressing the performance limitations of a previous LMBA design, which suffered from a narrow 3 dB output power back-off (OBO) range and single-frequency operation. Through a fundamental analysis, the OBO range was successfully doubled to 6 dB by re-biasing the auxiliary amplifier from Class AB to Class C. Subsequently, based on a review of state-of-the-art literature, an asymmetrical amplifier configuration was adopted, expanding the operational bandwidth and increasing the OBO range to 9 dB.

- **Phase 2: Transition to a Pseudo-Doherty LMBA (PD_LMBA) Architecture**
  To achieve ultra-wideband performance and a higher OBO range, the design transitioned to a more advanced PD_LMBA topology. This architecture inverts the roles of the main and auxiliary amplifiers to create a Doherty-like mechanism, which simplifies broadband design and inherently supports power asymmetry. By implementing this structure with new matching networks and a three-section branch-line coupler, the operational bandwidth was dramatically expanded to a **47% fractional bandwidth (2.75 GHz to 4.2 GHz)**, while achieving a 9-13 dB OBO range.

- **Phase 3: Novel Composite Right/Left-Handed (CRLH) Matching Network**
  A key innovation of this research was the design and integration of a novel network using a Composite Right/Left-Handed (CRLH) structure. This was proposed to solve the critical challenge of maintaining a linear phase response across the amplifier's wide bandwidth—a key requirement for PD_LMBAs. The CRLH network's inherent wideband properties, compact size, and linear phase response further expanded the simulated bandwidth to **55% (2.5 GHz - 4.4 GHz)**. The design was then translated from ideal lumped elements to a physically realizable microstrip equivalent for high-frequency operation.
- **Phase 4: Final Layout and Electromagnetic (EM) Simulation**
  The final phase focused on bridging the gap between schematic simulation and physical reality. The design was translated into a physical layout in Keysight ADS, where practical constraints were addressed. This included incorporating vendor-specific component models, accounting for the parasitic effects of soldering pads, and adhering to manufacturing tolerances. A final EM/schematic co-simulation was performed to validate the design, resulting in a robust, fabrication-ready layout with highly accurate performance predictions.

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3. Significant Results Achieved

The final EM-simulated design demonstrates state-of-the-art performance, achieving:

- **Fractional Bandwidth: 57%** (from 2.5 GHz to 4.5 GHz).
- **OBO Range: 9 to 12 dB** across the entire band.
- **Power Added Efficiency (PAE): 40% to 51%** at OBO and **48% to 60%** at saturation.

These results are highly competitive when compared with recently published research on similar amplifier topologies, validating the effectiveness of the design methodology and the innovative solutions implemented.

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4. Demonstrated Research Capabilities and Skills

This project provided a platform to develop and demonstrate a range of advanced engineering and research skills:

- **Systematic Problem-Solving:** Showcased a methodical approach to overcoming research challenges, effectively moving from independent analysis

to productive collaboration with supervisors and peers. Developed proficiency in leveraging academic literature to solve design impasses and implement innovative solutions.

- **Advanced RF Design and Optimization:** Mastered the complex trade-offs inherent in power amplifier design. Through extensive iterative simulation, skillfully balanced competing performance metrics (efficiency, bandwidth, and OBO range) to meet the project's stringent requirements.
- **Practical Engineering and Simulation Expertise:** Bridged the critical gap between theoretical simulation and practical implementation. Gained expertise in advanced design tools (Keysight ADS) and EM simulation, accounting for real-world factors like component parasitics and manufacturing tolerances to create a physically realizable and robust design.
- **Continuous Learning and Application of Theory:** Demonstrated a strong commitment to continuous learning by applying foundational knowledge in Microwave Engineering and Analog Electronics to solve complex, practical design problems. This hands-on approach successfully transformed abstract theoretical concepts into tangible engineering intuition.